

JEAN MARIE NTSAMA

Philosopher of Language - Cognitive Science Researcher
AI-Augmented Learning - Digital Literacies - Educational Innovation

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RESEARCH PROFILE

I am a transdisciplinary researcher bridging Philosophy of Language, Cognitive Science, and Computational AI for Education. My core expertise lies in modeling the dynamic interplay between language, thought, and technology. I am the founder of the Cognitive Trapeze Model, a computational and pedagogical framework that integrates AI as a mediating actor in learning environments. My research is deeply grounded in empirical field work within Priority Education Zones (ZEP) in Cameroon, where I develop and test offline, low-resource AI tools to support neurodiverse and multilingual learners.

Key Achievements

- **Theoretical Innovation:** Founder of the Cognitive Trapeze model - a computational framework formalizing language-thought co-development into a predictive Bayesian network.
- **Field Impact:** Founder of IA_4_ZEP, a field initiative reaching 160 learners and 60 teachers across 4 Cameroonian ZEP schools (2020-present).
- **Empirical Breakthrough:** First corpus-based evaluation of Whisper ASR models in real African ZEP classrooms (18 texts, 60+ learners, Apr-Jul 2026).
- **Public Sector Innovation:** Developer of SPS_DRES_AD, a public-sector educational data-management software deployed at MINESEC DRES Adamaoua.
- **Policy Impact:** Project Director of a GIZ-funded national project (2012) whose findings led the Cameroonian State to formalize volunteering activities.

CAREER PATH NOTE

A note on my unconventional trajectory: After completing my five-year multidisciplinary training at the Ecole Normale Supérieure of Yaounde (2008-2013), I signed a ten-year statutory service commitment to the Cameroonian State and was deployed to a remote Priority Education Zone (Adamaoua), where I served continuously. These years are not research gaps: they are the empirical foundation of my research. I defended my PhD on 27/07/2022 while teaching in this ZEP, and the Cognitive Trapeze model directly results from a decade of systematic classroom observation. Having completed my statutory service, I am now fully available for full-time research and European collaborations.

EDUCATION

- **27/07/2022:** PhD in Philosophy (Epistemology & Philosophy of Science) - University of Yaounde I. Registered 01/10/2012; successfully defended 27/07/2022 (grade: Tres Honorable). Thesis: Langage et pensée : acquisition, troubles et enseignements.
- **12/2013:** DIPES II (2nd-grade secondary teaching certification) - ENS Yaounde (10/2012-12/2013).
- **07/2011:** Master II in Philosophy of Science and Language - University of Yaounde I (10/2008-07/2011).
- **07/2011 & 07/2008:** BA in Modern French Letters & BA in Philosophy - University of Yaounde I / ENS (10/2009-07/2011 & 10/2005-07/2008).
- **07/2011:** DIPES I (1st-grade secondary teaching certification) - ENS Yaounde (10/2008-07/2011).

PROFESSIONAL EXPERIENCE

- **2020-Present:** Founder & Lead Researcher - IA_4_ZEP. AI-supported reading prototypes in 4 ZEP schools; ASR evaluation campaigns; formal partnership with MINESEC DRES.
- **10/2022-Present:** Teacher of Philosophy and French - Lycee Technique et Bilingue de Ngaoundere (ZEP). Differentiated pedagogy in 60+ learner classrooms; direct field base for action-research.
- **09/2016-04/2022:** Head of IT Unit & SIGIPES Relay Centre - MINESEC DRES Adamaoua. Developed/deployed SPS_DRES_AD; trained DRES managers; managed regional ICT infrastructure.
- **02/2015-07/2017:** Lecturer - IAI Cameroon (Ngaoundere). Courses: office production tools, network administration,

educational technologies.

- **2008-2012:** Elected leadership roles - AIESEC Yaounde. Project Director of the GIZ-funded 'Sur les traces du volontariat au Cameroun' (2012); Elected Vice-President for External Relations.

RESEARCH EXPEDITIONS & FIELD CAMPAIGNS

- **2026:** ASR evaluation campaign (Apr-Jul 2026), Adamaua ZEP schools: 18 curriculum texts, 60+ learners. Team: MINESEC inspectors, teachers. Role: lead investigator.

- **2020-2026:** Annual IA_4_ZEP data-collection campaigns in 4 ZEP schools (reading fluency, comprehension, audio recordings).

SELECTED PUBLICATIONS

- **[Preprint]** Ntsama, J. M., & Worsley, M. (2026). Evaluating Whisper ASR Engines in African Priority Education Zones. arXiv preprint (expected 09/2026). First corpus-based ASR benchmark in real African ZEP conditions.

- **[Article]** Ntsama, J. M. (2023). Web social, didactique et litteraties numeriques. Ifrikiya - Sciences du Langage (peer-reviewed).

- **[Monograph]** Ntsama, J. M. (2021). Etude des interrelations entre langage et pensee. EFUA. Theoretical basis of the Cognitive Trapeze.

- **[Monograph]** Ntsama, J. M. (2013). Problematique des troubles de langage et apport de la pedagogie curative. EUE. Introduces 'pedagogical curability'.

- **[Monograph]** Ntsama, J. M. (2013). Origine du langage : de la phylogenese a l'ontogenese. EUE.

OTHER RESEARCH OUTPUTS & INITIATIVES

- **Media:** NJM WebTV (2018-present): educational video channel and outreach platform for ZEP learners and digital literacies.

- **Software:** SPS_DRES_AD: public-sector educational data-management software deployed at MINESEC DRES Adamaua (2016-2022).

- **Dataset:** Open ASR corpus 'Texts 1-18' (to be deposited on Zenodo under CC-BY).

PRESENTATIONS, EVENTS, AWARDS & FUNDING

- **Talks:** Invited speaker, MINESEC regional pedagogical days and ICT training workshops (Adamaua, 2016-2026); AIESEC national/international seminars (2008-2012).

- **Organisation:** Organising Committee, AIESEC 'Leadership Building School' (2009); teacher workshops within IA_4_ZEP (2020-2026); regional ICT training sessions, MINESEC (2016-2022).

- **Awards:** AIESEC: Best Promotion Team Member; award for securing the GIZ grant (2011); project awards for each completed AIESEC project (2008-2012).

- **Funding:** GIZ project grant - EUR 4,573 (3,000,000 FCFA), obtained 2011 as Project Director. Findings contributed to State formalization of volunteering.

SUPERVISION, SKILLS & LANGUAGES

- **Mentoring:** Master's co-supervision in digital literacies (solicited 2026, University of Ngaoundere). Mentored 60 ZEP teachers via IA_4_ZEP workshops.

- **Technical:** Python (numpy, pandas, scikit-learn); ASR/LLM (Whisper, Hugging Face); mixed-methods research; R; NetLogo; AWS; Git/GitHub; Pydroid 3. Developing: Bayesian modelling, ABM (Mesa/NetLogo).

- **Languages:** French (native); English (C1, professional); Fulfulde (basic conversational, practised in ZEP).

This CV complements my MSCA Postdoctoral Fellowship 2026 application - project 'Cognitive Trapeze', hosted by the University of Copenhagen.